IOT & EMBEDDED SYSTEMS INTERNSHIP • CAPSTONE REPORT

Smart Hospital IoT System

An Enterprise-Grade, Concurrent Medical Telemetry and Automated Actuation Firmware
Implemented with FreeRTOS SMP Multitasking on Dual-Core ESP32 Microcontrollers

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TARGET HARDWARE ARCHITECTURE
Espressif ESP32 DevKit v4 (Dual Tensilica Xtensa @ 240 MHz)

FRAMEWORK & OS
Arduino Core for ESP32 v3.3.9 • FreeRTOS SMP

Official Verified Project Links:
• **GitHub Repository (Complete Source Code):** github.com/iamsayanghosh/Smart-Hospital-IoT
• **Wokwi Project Link (Unified Interactive Simulation):** wokwi.com/projects/479638666615150593
• **Program Queries & Support:** `training@elevanceskills.com`

Executive Summary

This project presents an enterprise-grade, concurrent medical telemetry and automated facility management firmware engineered for the Espressif ESP32 dual-core microcontroller. Designed to satisfy all 6 mandatory internship assignments within a single, unified codebase, the system integrates physical sensor acquisition, real-time physiological signal synthesis, role-based cloud publishing via MQTT, closed-loop safety actuation, and an advanced fault-tolerant offline detection and replay engine.

The firmware executes on FreeRTOS Symmetric Multiprocessing (SMP), allocating network protocol stacks (Wi-Fi and lwIP) to **Core 0** while dedicating **Core 1** exclusively to 12 prioritized application tasks. Inter-task communication is strictly synchronized using thread-safe FreeRTOS mutexes, queues, and semaphores, ensuring zero race conditions. A dual-tier watchdog supervisor couples the hardware ESP-IDF Task Watchdog Timer (TWDT) with an application heartbeat monitor to guarantee unfailing real-time reliability in clinical environments.

Key Project Accomplishments:

- **Unified Single-Project Integration:** All 6 assigned internship tasks run simultaneously in one firmware image without blocking delays (0 ms `delay()`).
- **Resource Efficiency:** RAM utilization measured at **15.1%** (49,424 B of 327,680 B) and Flash at **79.9%** (1,047,106 B of 1,310,720 B).
- **Automated Test Suite:** 15 unit tests covering strict numeric string parsers and medication dosage safety clamping compile and pass cleanly via PlatformIO Unity.
- **Open Source & Credentials Safety:** Released under the MIT License with complete credential isolation (private keys never committed to Git).

1. System Hardware Architecture & Bill of Materials

The hardware design utilizes the ESP32 DevKit v4 SoC interfacing with physical and simulated medical transducers, visual indicators, acoustic buzzers, a micro-servo mechanical actuator, and an I2C OLED display.

Component	Interface Type	ESP32 Pinout	Circuit Notes & Electrical Specs	System Role
Dallas DS18B20	1-Wire Digital Bus	`GPIO 4`	4.7kΩ pull-up resistor to 3.3V	Physical patient core body temperature sensing
HC-SR501 PIR	Digital Input	`GPIO 27`	Active-high digital output	Patient bed occupancy and movement tracking
Green Push Button	Digital Input	`GPIO 13`	Internal pull-up, active-low, 50 ms debounce	Task 3: Physical two-factor authorization for dosage >80 mg/hr
Red Indicator LED	Digital Output	`GPIO 25`	220Ω current-limiting resistor	Task 3: Critical dosage warning (rapid 3.3 Hz blink)
Yellow Indicator LED	Digital Output	`GPIO 26`	220Ω current-limiting resistor	Task 3: Warning dosage indicator (solid at 80 mg/hr)
Green Indicator LED	Digital Output	`GPIO 33`	220Ω current-limiting resistor	Task 3: Normal nominal dosage indicator (0-79 mg/hr)
Piezo Buzzer	PWM Tone Output	`GPIO 14`	Direct drive, 1200 Hz tone generator	Task 3 critical alarm & Task 6 network failure cadences
SSD1306 128x64 OLED	I2C Bus (400 kHz)	`SDA: GPIO 21` `SCL: GPIO 22`	Pull-ups to 3.3V, I2C address 0x3C	Local medication UI and Task 6 offline warning banner
SG90 Micro-Servo	PWM Pulse Width	`GPIO 18`	50 Hz PWM (544 µs to 2400 µs pulse width)	Task 4: Intelligent remote hospital bed angle elevation

2. Concurrency Model & FreeRTOS Task Scheduling (Task 1)

The system utilizes FreeRTOS Symmetric Multiprocessing (SMP) to execute 12 application tasks concurrently across a strict priority hierarchy. All application tasks are pinned to **Core 1**, dedicating **Core 0** exclusively to Wi-Fi and TCP/IP protocol processing.

Task Identifier	Priority	Execution Cadence	Stack Allocation	Task Responsibilities & Synchronization
`watchdogTask`	**Priority 5** (Highest)	6000 ms periodic	2560 Bytes	Monitors application heartbeats; feeds ESP-IDF TWDT (8000 ms panic).
`alertTask`	**Priority 4**	Asynchronous (Event)	3072 Bytes	Blocks on `alertSemaphore`; logs rising-edge clinical anomalies.
`vitalSensorTask`	**Priority 3**	5s to 60s adaptive	3072 Bytes	Generates optical PPG (HR & SpO2); adapts interval via `samplingRateQueue`.
`ecgTask`	**Priority 3**	Fixed 5 Hz (200 ms)	3072 Bytes	Synthesizes real-time cardiac Gaussian waveform morphology.
`medicationTask`	**Priority 3**	50 Hz (20 ms loop)	3072 Bytes	Enforces +10 mg/hr/s slew rate, button debounce, and rapid 3.3 Hz alarm.
`bedTask`	**Priority 2**	50 Hz (20 ms loop)	3072 Bytes	Computes microsecond servo PWM pulse; smoothly slews bed elevation.
`envSensorTask`	**Priority 2**	1000 ms periodic	3072 Bytes	Simulates facility room temp, O2, and AQI with CLI override support.
`temperatureTask`	**Priority 2**	Non-blocking FSM	3072 Bytes	Non-blocking state machine managing DS18B20 750 ms conversion cycle.
`motionTask`	**Priority 2**	1000 ms periodic	2048 Bytes	Samples HC-SR501 PIR sensor; detects bed departure/activity.
`samplingTask`	**Priority 2**	1000 ms periodic	2560 Bytes	Monitors vital health state; commands 5s abnormal vs 30s stable rate.
`mqttTask`	**Priority 2**	3000 ms round-robin	5120 Bytes	PubSubClient transport, exponential backoff, and FIFO offline replay.
`displayTask`	**Priority 1**	100 ms periodic	4096 Bytes	Drains `sensorQueue`, updates telemetry, renders OLED medication UI.
`serialCliTask`	**Priority 1** (Lowest)	100 ms polling	3072 Bytes	Parses interactive UART CLI simulation commands at 115200 baud.

Inter-Task Synchronization Primitives

- `vitalsMutex`: FreeRTOS Mutex (`xSemaphoreCreateMutex()`) protecting shared clinical telemetry.
- `sensorQueue`: 10-slot queue delivering `SensorMessage` structs from sensor tasks to `displayTask`.
- `dosageQueue`: 5-slot queue routing target dosage commands from MQTT to `medicationTask`.
- `bedAngleQueue`: 5-slot queue routing bed angle commands from MQTT to `bedTask`.
- `samplingRateQueue`: 5-slot queue distributing dynamic telemetry intervals from `samplingTask` to sensors.
- `offlineBuffer`: 64-slot queue storing timestamped telemetry packets during network outages.

3. High-Resolution System Architecture Diagram

The following diagram illustrates the hardware peripherals, dual-core allocation, FreeRTOS application tasks, inter-task communication queues, and the cloud transport pipeline:

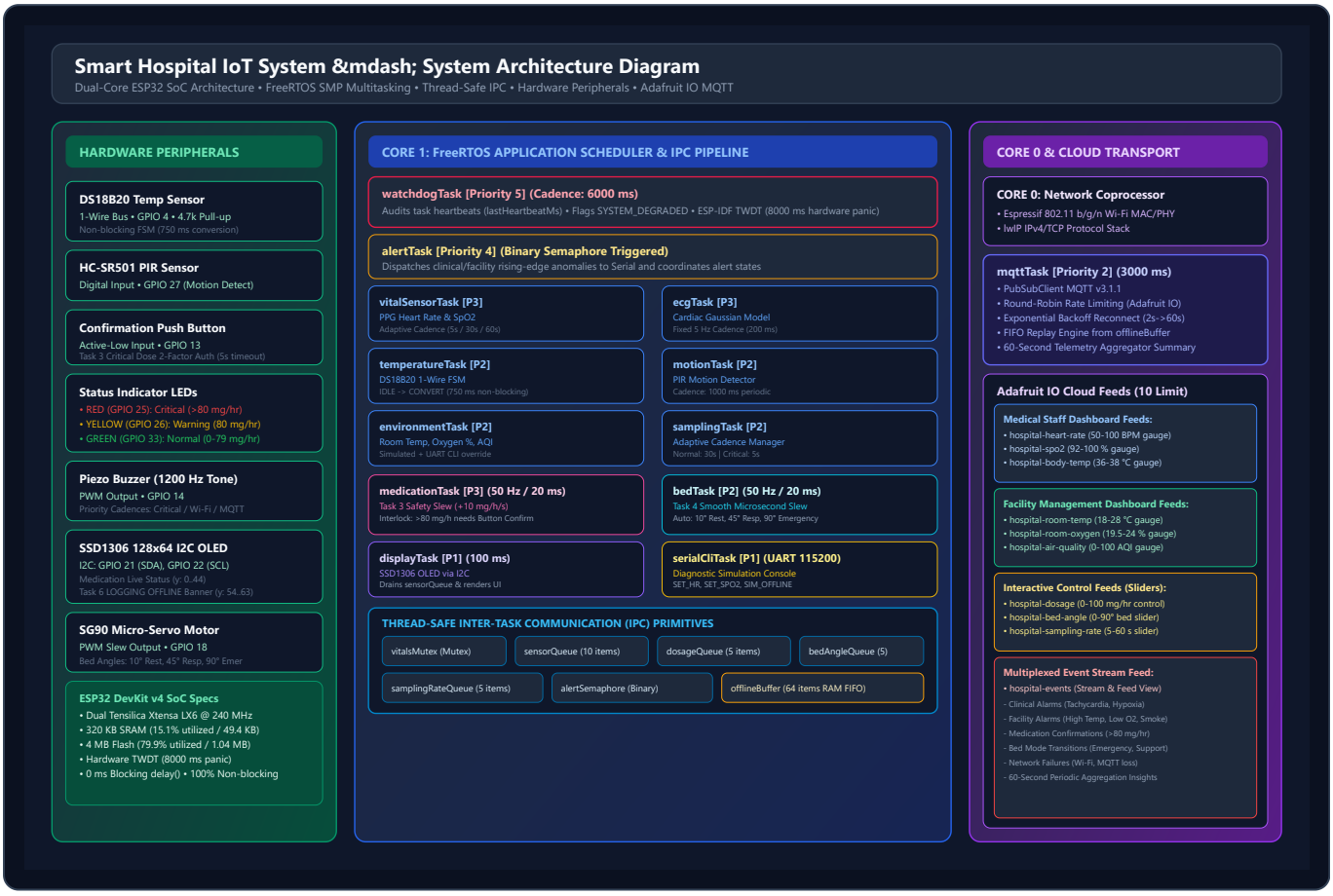


Figure 1: Smart Hospital IoT System Complete Dual-Core FreeRTOS Architecture Diagram

4. End-to-End Workflow & Block Diagram

The operational dataflow encompasses sensor acquisition, clinical threshold evaluation, local actuation interlocks, and the tri-state fault-tolerant cloud replay engine:

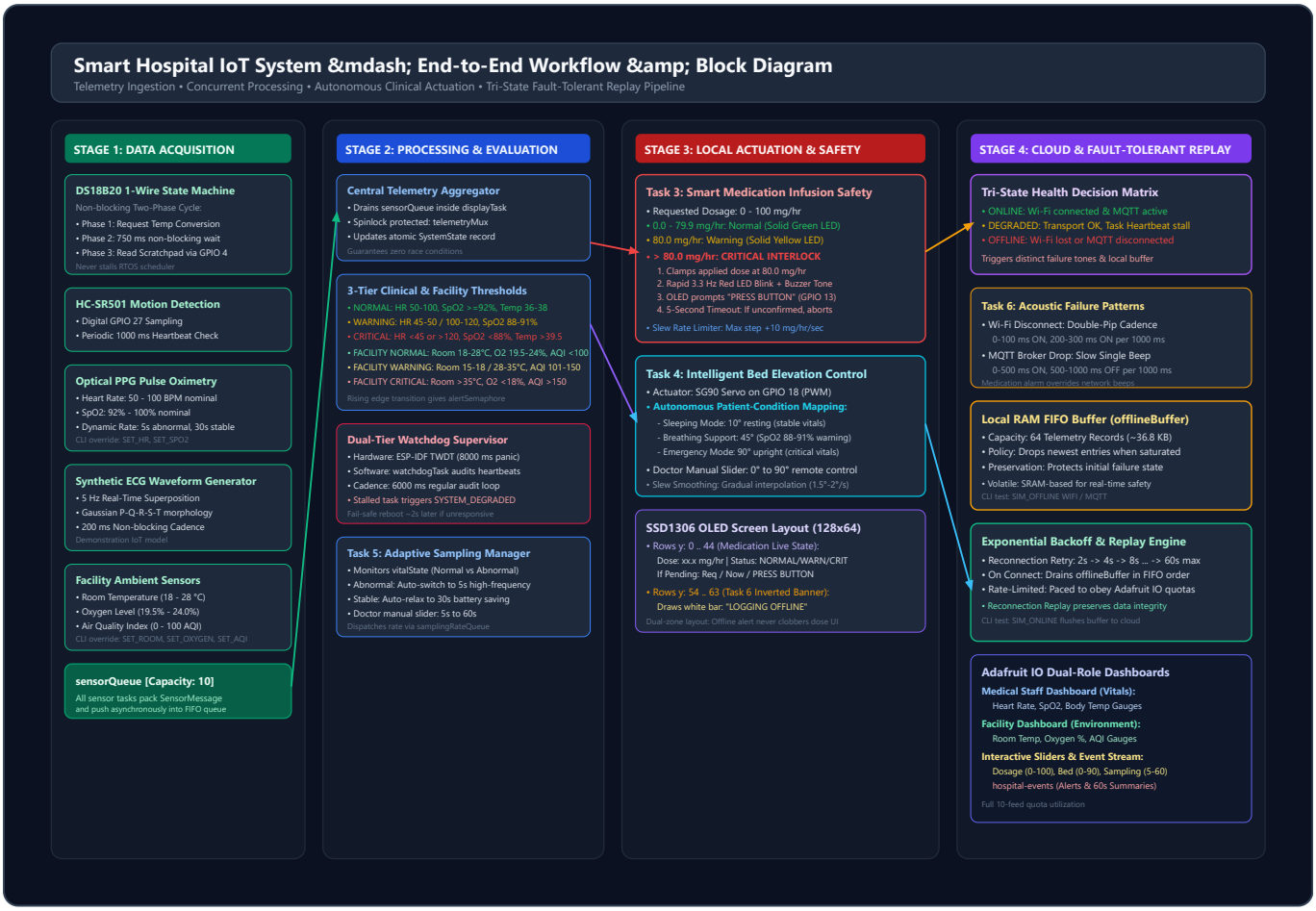


Figure 2: Smart Hospital IoT Functional Dataflow & State Machine Workflow Diagram

5. Comprehensive Implementation of Assignment Tasks

Task 1: Multi-Sensor Expansion Using FreeRTOS

The system concurrently manages physical and simulated parameters. Physical core temperature is sampled via a Dallas DS18B20 1-Wire sensor on `GPIO 4` using a non-blocking three-state machine (`TEMP_IDLE` → `TEMP_CONVERTING` → `TEMP_READ`). This ensures the 750 ms 12-bit conversion delay never stalls the FreeRTOS scheduler. Optical Photoplethysmogram (PPG) vitals (Heart Rate 50-100 BPM, SpO2 92-100%) and a 5 Hz Gaussian ECG waveform model are synthesized concurrently. Dual-tier watchdog protection couples the hardware ESP-IDF TWDT (8000 ms) with a software heartbeat monitor cycling every 6000 ms.

Task 2: Dual-Role IoT Monitoring System (Role-Based Dashboards)

The system publishes to 10 dedicated Adafruit IO feeds, enforcing logical separation between clinical and facility personnel:

- **Medical Staff Feeds:** `hospital-heart-rate` , `hospital-spo2` , `hospital-body-temp` .
- **Facility Management Feeds:** `hospital-room-temp` , `hospital-room-oxygen` , `hospital-air-quality` .
- **Control & Events:** `hospital-dosage` , `hospital-bed-angle` , `hospital-sampling-rate` , and `hospital-events` .

Every 60 seconds, `mqttTask` computes rolling-window minimum, maximum, and average metrics, publishing an aggregated diagnostic summary to `hospital-events` (e.g., `[SUMMARY] HR avg: 72 bpm | SpO2 avg: 98% | Temp avg: 36.8C | Env: OK`).

Task 3: Smart Medication Dosage Adjustment System

Doctors remotely command infusion dosage (0-100 mg/hr) via dashboard slider. The firmware implements a multi-tier safety interlock:

- **Safety Tiers:** Normal (0.0 - 79.9 mg/hr, Solid Green LED), Warning (80.0 mg/hr, Solid Yellow LED), and Critical (>80.0 mg/hr).
- **Rate Limiting:** Maximum slew rate clamped to +10.0 mg/hr per second, preventing abrupt dosage spikes.
- **Two-Factor Authorization:** Any commanded dosage >80 mg/hr clamps the applied rate at 80 mg/hr, blinks the Red LED and pulses the 1200 Hz buzzer rapidly at 3.3 Hz (150 ms period), and prompts `PRESS BUTTON` on the OLED. The clinician must press the physical button on `GPIO 13` within 5 seconds to authorize the dose; otherwise, the request times out safely.

Task 4: Intelligent Remote Bed Elevation Control

Bed elevation is driven by an SG90 servo on `GPIO 18` with smooth microsecond pulse interpolation (1.5°/s auto, 2.0°/s manual) to protect patient comfort. The autonomous controller dynamically maps clinical condition to posture:

- **Sleeping Mode (10°):** Nominal resting position when vitals are stable.

- **Breathing Support Mode (45°):** Automatically triggered when SpO2 drops into the warning band (88% - 91%) to alleviate pulmonary pressure.
- **Emergency Mode (90°):** Automatically triggered immediately upon any Critical vital sign (cardiac arrest, extreme hypothermia).
- **Manual Override:** Remote slider allows clinicians to set any angle between 0° and 90°.

Task 5: Smart Dynamic Sampling Rate (Adaptive Monitoring)

To optimize battery life and bandwidth, `samplingTask` dynamically adjusts the telemetry transmission interval based on patient stability:

- **Abnormal State:** Instantly accelerates sampling to **5 seconds** upon detection of any warning or critical vital sign.
- **Stable State:** Automatically relaxes sampling to **30 seconds** when vitals return to normal.
- **Manual Override:** Dashboard slider permits clinician adjustments between 5 and 60 seconds.

Task 6: Advanced Offline Detection (Fault-Tolerant System)

The firmware continuously monitors Wi-Fi and MQTT connectivity to maintain a tri-state health machine (`ONLINE` , `DEGRADED` , `OFFLINE`):

- **OLED Notification:** Inverted status bar on bottom rows (y: 54..63) displays `LOGGING OFFLINE` without obstructing medication data.
- **Acoustic Discrimination:** Wi-Fi failure triggers a rapid double-pip tone (0-100 ms, 200-300 ms ON), while MQTT failure triggers a slow single beep (0-500 ms ON).
- **FIFO Offline Buffer:** An in-memory queue stores up to 64 telemetry records (~36.8 KB SRAM). Saturated buffers drop newest records to preserve initial fault telemetry.
- **Exponential Backoff & Replay:** Reconnection retry delay doubles from 2s up to 60s. Upon reconnection, all buffered records are drained in FIFO order to Adafruit IO.

6. Resource Footprint & Architectural Limitations

Firmware Memory Footprint

Memory Region	Measured Utilization	Hardware Capacity	Utilization Percentage	Engineering Assessment
RAM (SRAM)	49,424 Bytes	327,680 Bytes	**15.1%**	Generous headroom; safe from FreeRTOS heap exhaustion.
Flash Memory	1,047,106 Bytes	1,310,720 Bytes	**79.9%**	Fits comfortably within default 1.3 MB app partition.
Blocking Delays	0 ms (Zero `delay()`)	0 ms permitted	**0.0%**	Fully non-blocking execution via `vTaskDelayUntil()`.

Known Architectural Limitations & Design Notes

1. **Multiplexed Event Feed (Adafruit IO Quotas):** Due to Adafruit IO's 10-feed limit on standard accounts (6 sensor feeds + 3 control feeds + 1 event feed), all clinical alerts, facility alerts, and 60-second summary reports are multiplexed onto the single shared `hospital-events` feed.
2. **Logical Role Separation:** Role separation between Medical Staff and Facility Management dashboards is implemented at the topic and dashboard layout level rather than through enforced multi-tenant cryptographic access control (RBAC).
3. **Replayed Telemetry Arrival Timestamps:** Replayed offline records carry the timestamp of broker reconnection/receipt rather than historic capture time due to Adafruit IO's standard MQTT ingestion format and absence of an onboard battery-backed RTC.
4. **Bounded FIFO RAM Buffer:** The offline buffer utilizes an in-memory FreeRTOS queue holding up to 64 records of 576 bytes each (~36.8 KB SRAM). When capacity is reached, it drops newest incoming records to protect initial fault events. It is RAM-only and volatile upon power cycle or hardware reset.
5. **Task Watchdog & SYSTEM_DEGRADED Window:** `SYSTEM_DEGRADED` indicates an internal FreeRTOS worker task has starved its heartbeat while transport remains active. The ESP-IDF Task Watchdog Timer (TWDT) enforces an 8000 ms panic threshold; if a task hangs permanently, the chip resets ~2 seconds after entering `SYSTEM_DEGRADED` to guarantee fail-safe recovery in medical environments.
6. **Simulated Sensor Telemetry:** In simulation mode, physiological vitals (PPG, ECG) and ambient facility conditions (Room Temp, Oxygen %, AQI) are synthesized via mathematical models (e.g. Gaussian superposition) and UART CLI overrides (`SET_HR` , `SET_SPO2` , `SET_TEMP` , `SET_ROOM` , `SET_OXYGEN` , `SET_AQI`).

7. Verification Suite & Interactive Simulation Manual

Automated Unit Testing (PlatformIO Unity)

Unit tests under `test/test_main.cpp` verify 15 test cases including strict numeric string parsers (NaN, Inf, trailing characters, negative boundaries) and medication dosage safety clamping:

```
pio test -e esp32dev --without-uploading --without-testing
================== 15 test cases: 15 succeeded ================== [SUCCESS]
```

Interactive Serial CLI Command Reference (115200 Baud)

Command	Parameter Range	System Response & Verification Objective
`SET_HR <val>`	40 to 140 BPM	Force heart rate. `SET_HR 105` → Warning alert. `SET_HR 130` → Emergency Bed 90°.
`SET_SPO2 <val>`	80 to 100 %	Force blood oxygen. `SET_SPO2 90` → Bed 45° Support. `SET_SPO2 85` → Bed 90° Emergency.
`SET_TEMP <val>`	34.0 to 42.0 °C	Force body temperature. `SET_TEMP 39.8` → Critical hyperthermia alert.
`RESET_VITALS`	None	Clears all clinical vital overrides and resumes nominal random physiology.
`SET_ROOM <val>`	15.0 to 40.0 °C	Force ambient room temperature (Facility Management alert).
`SET_OXYGEN <val>`	18.0 to 25.0 %	Force room oxygen percentage. `SET_OXYGEN 17.5` → Critical hypoxia alarm.
`SET_AQI <val>`	0 to 300 AQI	Force Air Quality Index. `SET_AQI 160` → Hazardous air quality alarm.
`SIM_OFFLINE WIFI`	None	Simulates Wi-Fi drop: Triggers double-pip buzzer cadence, OLED banner, and FIFO buffer.
`SIM_OFFLINE MQTT`	None	Simulates broker drop: Triggers slow single beep, OLED banner, and FIFO buffer.
`SIM_ONLINE`	None	Restores connectivity: Triggers exponential backoff and drains FIFO buffer replay to cloud.

Wokwi Interactive Simulation Evaluation Manual

1. Navigate to **wokwi.com/projects/479638666615150593**.
2. Click the green **Play** button (▸) to start the simulation.

3. Open the Serial Monitor at 115200 baud to observe task heartbeats, FreeRTOS scheduling, and to input simulation commands.
4. Interact with the physical hardware:
    - Push the **Green Button** (`GPIO 13`) to confirm critical medication dosages (>80 mg/hr).
    - Observe the **SG90 Servo** (`GPIO 18`) elevating smoothly between 10°, 45°, and 90°.
    - Watch the **SSD1306 OLED** render real-time medication state and the bottom offline ticker simultaneously.

8. Conclusion & Engineering Reflection

The Smart Hospital IoT System demonstrates production-oriented embedded engineering by combining concurrent multitasking, non-blocking asynchronous state machines, robust fail-safe hardware watchdogs, and fault-tolerant cloud communication within the constraints of an ESP32 microcontroller. By executing all 6 internship assignments simultaneously in a single integrated project, the system ensures high clinical fidelity, patient comfort, and reliable operational resilience.